

Technology Stack

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Technology Stack:

The Smart Lender Loan Eligibility Prediction System uses a combination of frontend, backend, machine learning, and deployment technologies to build an efficient and user-friendly application. The selected technology stack ensures scalability, maintainability, and accurate loan prediction by providing a responsive user interface, efficient data processing, secure model deployment, and effective management of project resources.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML, CSS, Bootstrap	Used to create a responsive and user-friendly interface for entering loan details and viewing results.
2	Backend / Server-Side	Python, Flask	Provides fast processing and integrates easily with machine learning models.

3	Database / Data Storage	CSV Dataset	Simple and suitable for storing and processing loan applicant data for machine learning.
4	Cloud / Hosting / Deployment	Local Host / Render / Heroku	Easy and cost-effective deployment of the web application.
5	Version Control & CI/CD	GitHub	Enables code management, version tracking, and collaboration.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Joblib	Used for data preprocessing, machine learning model development, and model storage.